Bhavana Kondeti

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SUMMARY

Full-Stack Software Engineer with experience building scalable enterprise SaaS platforms using React, Java, PHP, and Python. Strong collaborator in Agile teams, contributing to backend modernization, performance optimization, and user-facing feature development. Research exposure to AI-driven software testing (LLMs, RAG) with a strong drive to explore emerging technologies and solve complex engineering challenges.

TECHNICAL SKILLS

Backend & APIs:	PHP, Python, Java, JavaScript, SQL, RESTful APIs, JWT/OAuth2, Microservices, Data Validation, Spring Boot, Spring WebFlux (Reactive Programming), Error Handling, Rate Limiting, RAG, Vector Retrieval, RAG pipelines
Frontend:	React.js, Next.js, TypeScript, JavaScript, Datastar
Databases & Data:	PostgreSQL, MySQL, MongoDB, Firebase, Database Design, Query Optimization, Indexing
Cloud & DevOps:	AWS (EC2, S3, RDS, Lambda, API Gateway, IAM), Docker, CI/CD Pipelines, GitHub Actions, Cloud Deployment, Auto Scaling
Testing & Quality:	Vitest, Unit Testing, Integration Testing, API Testing, TDD, Regression Testing, Test Automation
Architecture & Engineering:	Scalable Systems, System Design, Distributed Systems (Foundations), Performance Optimization, Fault Tolerance
Tools &Practices:	Git, Gitlab, Agile, Code Reviews, Release Management

EXPERIENCE

Paycom | Software Engineer | USA — **Jan 2025 – Present**

- Collaborated with product and stakeholder teams to translate enterprise client requirements into scalable full-stack solutions serving **thousands of users** across core HCM workflows.
- Led development and modernization of full-stack applications using **React (TSX)**, including a full rewrite of a legacy frontend to React with improved architecture and test coverage.
- Delivered **massive performance improvements (up to 3000×)** by optimizing SQL queries and refactoring data-access patterns in critical user paths
- Engineered automation to detect and escalate stalled expense approvals, reducing manual intervention and improving operational efficiency.
- Researched and prototyped a server-enhanced frontend POC to assess performance and scalability improvements for the expense platform, reducing frontend overhead in high-traffic workflows.
- Improved platform stability by **rapidly debugging client-reported issues**, delivering accurate fixes or workarounds, and enhancing search, sorting performance, and **i18n** support.

Paycom | Software Engineer Intern | USA — **May 2024 – Aug 2024**

- Designed and implemented a **centralized Import Center architecture**, introducing a modular, configuration-driven framework to standardize enterprise data ingestion across multiple product domains, reducing code duplication by 40%.
- Contributed to a **layered backend architecture (Controller → Service → Validation → Repository)** to enforce separation of concerns and improve maintainability and testability.
- Integrated **RabbitMQ-based asynchronous processing** to decouple file ingestion from validation and persistence workflows, improving scalability and preventing request blocking during large dataset uploads.
- Architected a **pluggable validation pipeline**, enabling configurable business rules and dynamic field-level validations for secure and compliant enterprise data ingestion.
- Developed RESTful APIs and integrated them with React-based frontend workflows, enabling real-time validation feedback and high-volume file upload handling.
- Optimized database access patterns and indexing strategies to support large-scale data processing with improved throughput and reliability.

University of Texas at San Antonio – Research Assistant **Aug2023 – Jan 2026**

- **Built EvaDroid,** an AI-powered Android UI exploration platform combining LLMs, RAG, and vector search to autonomously test mobile applications and outperform random and rule-based strategies.
- **Designed and implemented a multi-agent orchestration system**, coordinating VisionAgent (UI parsing), RAG Agent (contextual retrieval), DecisionAgent (LLM reasoning), and ExecutionAgent (ADB automation) to enable end-to-end autonomous workflows.
- **Developed backend services in Python** to process UI dumps, extract actionable elements, generate embeddings, and persist state–action pairs in a vector database for long-term memory and contextual reasoning.
- **Deployed and managed infrastructure on AWS (EC2, S3, IAM)** to support large-scale UI log storage, model interaction workflows, and experimental benchmarking pipelines.
- **Engineered a feedback-driven reward modelling framework**, evaluating actions based on UI coverage gains, newly discovered elements, and workflow progression to improve decision quality across sessions.
- **Integrated LLM pipelines with retrieval augmentation,** embedding UI states and historical interactions to enable adaptive decision-making across long-running exploration runs.
- **Led experimental evaluation and benchmarking**, comparing LLM-only, RAG-enhanced, and random strategies, demonstrating improved coverage, efficiency, and stability; published as **First Author (ICSE NIER).**

Apxor Technology Solution | Software Developer **Mar 2022 - Jan 2023**

- Collaborated with Product Managers and Engineering teams to design, develop, and ship high-impact features and enhancements, contributing to a **40% increase in user conversion rates** through performance improvements and UX refinements.
- Increased frontend test coverage to **85% using Vitest**, implementing structured unit and integration testing strategies, leading to a **45% reduction in production defects** and improved release confidence.
- Designed and maintained internal **test harnesses and demo applications** using React.js, Next.js, Angular, Django, and Vue, enabling comprehensive cross-framework testing, integration validation, and rapid feature experimentation.
- Strengthened system reliability and integration quality by improving input validation and error handling across frontend and backend workflows, supporting enterprise API integrations, and mentoring interns through technical walkthroughs and architecture guidance to accelerate team productivity.

PROJECTS

Serverless Full-Stack Application on AWS

- Architected and deployed a serverless cloud application leveraging **AWS Lambda, API Gateway, RDS (PostgreSQL), S3, and Amplify**, eliminating infrastructure management overhead.
- Configured IAM roles and least-privilege policies to secure Lambda-to-RDS communication and API access.
- Implemented optimized PostgreSQL queries and connection handling to support scalable request processing.
- Hosted and delivered a production ready React frontend via S3, integrating with backend APIs through API Gateway.
- Automated deployment workflows using Amplify CLI for streamlined full-stack provisioning.

EDUCATION

University of Texas at San Antonio (UTSA)| M.S Computer Science
Gayatri Vidya Parishad College of Engineering, India| B.Tech Computer Science

ACHIEVMENTS

Founded **TraditionOnWay**, an **IIM-B NSRCEL-selected startup incubated at IIM-V**, focused on delivering authentic products directly from manufacturers to customers, ranging from clothing to foods. Recognized among **the top 200 startups in the Women Startup Program out of 10,000 submissions,** the company has built a strong customer base and strong impactful sales.